

How the NERSC User Community Connects

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Abstract—High-performance computing centers, such as NERSC, serve thousands of projects and users across many science domains and funding offices, yet staff have few systematic ways to see how these communities form, persist, and connect for planning and outreach. As an exploratory step toward understanding whether allocation data can surface actionable insights, 2020–2026 Perlmutter allocation records are turned into yearly project and user networks and Louvain community detection, cross-office bridge detection via weighted betweenness, and user-level liaison ranking are applied. Null-model and matched-popularity tests confirm the structure is real and actionable: communities stay above the 95% null modularity band every year; a small set of key projects, about 10%, carries 66–74% of cross-office connectivity; and the top 200 liaison candidates reach 69% of active users, enabling efficient outreach. To our knowledge, this is the first network analysis of the community of an HPC facility.

Index Terms—network analysis, community detection, HPC allocations, betweenness centrality, ETL pipeline, user engagement

I. INTRODUCTION

High-performance computing (HPC) centers such as the National Energy Research Scientific Computing Center (NERSC) support thousands of projects and users spanning many scientific domains and DOE funding offices. Understanding how these users engage and how new engagements form would help staff plan, yet finding ways to determine these features beyond one-on-one conversations is heavily restrictive. This research investigates whether allocation data reveals actionable structure: whether communities are stable across offices and years, whether particular projects drive the connections between them, and whether individual users can be flagged as likely liaison candidates. Our contribution is a network analysis [5] of NERSC allocation records for community understanding, spanning project-level community and bridge structure through user-level liaison discovery.

II. DATA AND NETWORK CONSTRUCTION

Allocation and request records are processed from NERSC’s system, spanning 2020 through April 2026, with a repro-

ducible extract–clean–validate–load pipeline that standardizes identifiers and stores tables in DuckDB [7]. Each year has 8,000–13,000 active users and 850–1,200 active projects across six DOE program offices; the user base is early-career heavy (~60% students and postdocs), and 41% of the 2020 cohort is active in 2026.

For each year two graphs [1] are built (Fig. 1): a weighted project–project graph, where an edge joins two projects sharing at least one user and is weighted by the number of shared users, and a bipartite user–project graph used for liaison analysis.

III. METHODS AND VALIDATION

Louvain community detection [2] and weight-aware modularity Q [3] are run on each yearly project graph using NetworkX [1] and pose three questions:

Are the detected communities likely describing real communities, and are they stable across years? Observed modularity is compared to a degree-preserving edge-swap null [8] of 100 rewired graphs per year and its 95% band, and year-to-year membership overlap is compared using Jaccard to a size-preserving membership-shuffle null of 30 shuffles per year.

Is cross-office connectivity concentrated in a few bridge projects, beyond what their degree explains? A project is a bridge in a year when its weighted betweenness [4] is in the top 10% and it touches at least two offices; a persistent bridge meets this every year, and persistent-bridge removal is compared to 200 matched non-bridge removals.

Can a small shortlist of users be identified that is likely to reach most of the active community? Liaison candidates are ranked by scoring active users on tenure, projects per year, largest project, and bipartite betweenness.

IV. RESULTS

A. Community structure

The analysis suggests communities are real and structured: observed modularity stays above the 95% null band every year (Fig. 2a), and each year has a few very large communities with a long tail (Fig. 2b). The biggest communities are more office-mixed than typical (Fig. 2c), remain recognizable through

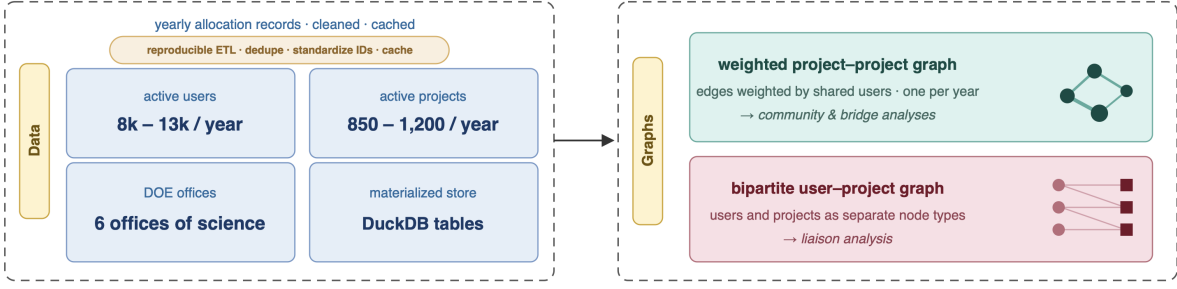


Fig. 1: From allocation records to a weighted project–project graph and a bipartite user–project graph.

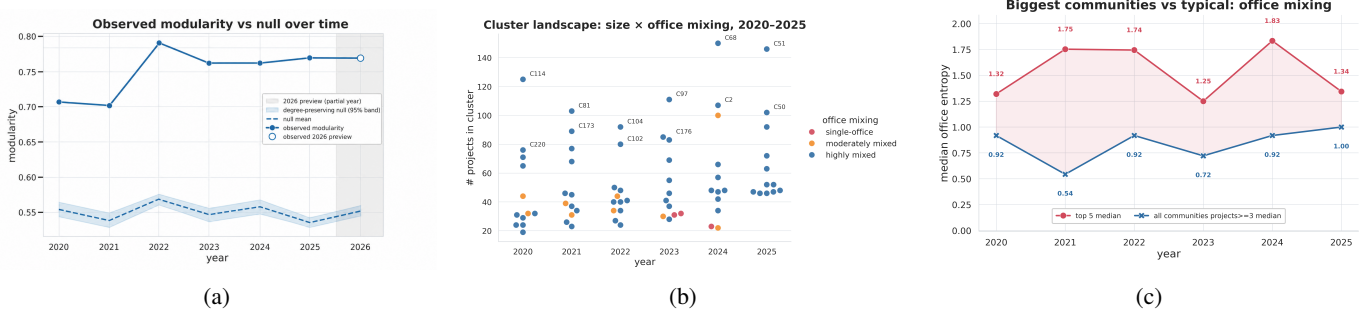


Fig. 2: Community structure, 2020–2025: (a) modularity vs. the 95% null band; (b) cluster size vs. office-mix entropy [6]; (c) biggest communities are more office-mixed than typical.

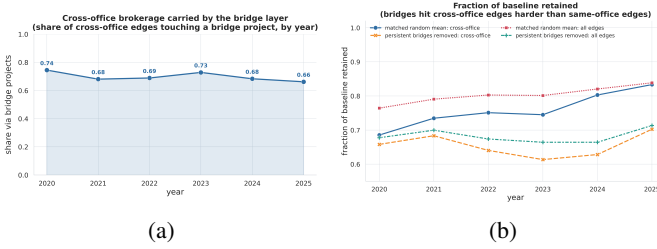


Fig. 3: Key bridges: (a) bridges touch 66–74% of cross-office edges each year; (b) removing persistent bridges drops more cross-office edges than matched removals.

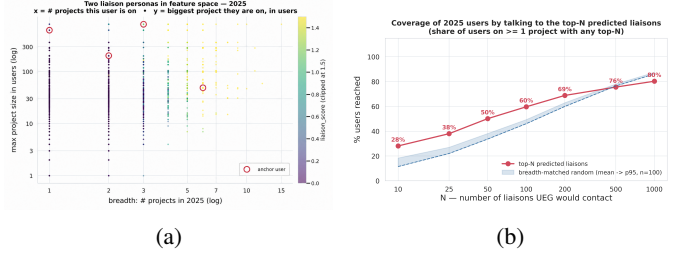


Fig. 4: Liaison discovery, 2025: (a) two personas in feature space; (b) coverage of active users by the top-ranked candidates.

2025 while the shuffled baseline is near zero, and tend to have higher GPU share, so project-sharing is non-random, and communities stay identifiable across years.

B. Key bridges

Cross-office connectivity appears concentrated in a small bridge layer: about 10% of projects meet the bridge definition in a year yet carry 66–74% of all cross-office edges (Fig. 3a), and 30 are persistent bridges. Removing them hurts cross-office connectivity more than matched removals (Fig. 3b), so this is not explained by degree alone.

C. Liaison discovery

A short, data-derived shortlist enables reaching most active users: the four anchor users separate into two liaison types (large-project-centered and multi-project bridge, Fig. 4a), and

the top 50, 100, and 200 candidates reach about 50%, 60%, and 69% of active users (Fig. 4b), enough for data-targeted outreach.

V. CONCLUSION AND FUTURE WORK

Network analysis turns allocation records into a decision-support view of how projects, users, and programs connect over time. Persistent bridge projects play an outsized role in cross-office connectivity, and a short liaison shortlist can reach most active users. Next steps are validating the liaison shortlist against support tickets, testing alternative community resolutions, and packaging the workflow into a dashboard. Code and a synthetic-data artifact are publicly available at <https://github.com/sumaiyaazad/How-NERSC-Connects>.

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